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Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

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Title

Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

Corresponding author

Mia Bierbaum, mia.bierbaum@hdr.mq.edu.au,

Australian Institute of Health Innovation, 75 Talavera Rd, Macquarie University, NSW, Australia

Authors

[1] Bierbaum, Mia

[1] Rapport, Frances

[1] Arnolda, Gaston

[1] Tran, Yvonne

[1] Nic Giolla Easpaig, Brona

[1] Ludlow, Kristiana

[1] Braithwaite, Jeffrey

Affiliations

[1] Australian Institute of Heath Innovation, Macquarie University, NSW, Australia

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- Oncology
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Abstract:

Introduction: Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making. CPG adherent care is associated with improved patient survival outcomes, however, adherence rates are low across some cancer streams in Australia. Greater understanding of specific barriers to adherence to cancer treatment CPGs is warranted to inform future implementation strategies.

This paper presents the protocol for a systematic review that aims to determine cancer treatment CPG adherence rates in Australia across a variety of common cancers, and identify any factors associated with adherence to those CPGs, as well as any associations between CPG adherence and patient outcomes.

Methods and analysis: Five databases will be searched, Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science, for eligible studies evaluating adherence

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rates to cancer treatment CPGs in Australia, published between 2000-2021. Five experienced reviewers will screen the abstracts according to predetermined inclusion criteria. The full text of eligible studies will then be reviewed and assessed for inclusion. All included studies will be assessed for quality and risk of bias. Data will be extracted using a predefined data extraction template. The frequency or rate of adherence to CPGs, factors associated with adherence to those CPGs, and any reported patient outcome rates (e.g., relative risk ratios or 5-year survival rates) associated with adherence to CPGs, will be described. If applicable, a pooled estimate of the rate of adherence will be calculated by conducting a random effects meta-analysis.

Ethics and dissemination: Ethics approval will not be required, as this review will present results from other published studies. Results from this study will form part of the lead author's doctoral dissertation. Results will also be published in a peer-reviewed journal, presented at conferences, and in other academic presentations.

Prospero number: CRD42020222962

Strengths and limitations of this study

- This systematic review protocol describes a planned review that will methodically identify primary research studies that estimate the rates of adherence to cancer treatment CPGs in Australia.

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- The review will highlight key factors associated with cancer treatment CPG adherence in Australia to inform future implementation strategies of cancer treatment CPGs.
- An internationally recognised methodology for the collection, extraction, and synthesis of data used in this systematic review will be transparently presented to enable replication.
- The review will be limited to studies published in English, between 2000 to 2021 to ensure the CPG adherence data is current.
- This study will also be limited to cancer treatment specific CPGs, excluding CPGs that relate to other elements of the cancer care journey such as pain management, and psychosocial care.

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Introduction

Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making,[1] and are designed to reduce clinical variation.[2 3] There is growing evidence that CPG adherent cancer care is linked to improved patient survival rates.[4-8] Despite this, adherence to cancer treatment CPGs in Australia is often reported to be low across a variety of cancer streams.[4 6 9-14] Variation from CPG recommendations in some instances can be reasonably justified to cater for individual patient preferences and needs.[5 15] However, little is known about the factors that influence these decisions.[16] Low uptake and adherence to cancer treatment CPGs may also reflect poor quality CPGs, or poor implementation of CPGs. Greater understanding of specific barriers to adherence to cancer treatment CPGs is warranted, to inform future implementation strategies.

There are a multitude of internal (attitudinal) and external (organisational and structural) barriers to cancer treatment CPG adherence. These include clinician concern that CPGs are biased or oversimplified, a lack of clinician awareness of, or agreement with, CPG recommendations, inertia,[15 17] and concerns about the side effects of CPG recommended treatment.[15] Similarly, patient concerns about side effects or discomfort related to treatments, and access to treatment, have also been identified as barriers to CPG adherence.[18] Collaboration within multidisciplinary teams during treatment decision-making has been found to influence adherence to CPG recommendations,[19-21] as has patient access to services,[4] the location of the

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treating hospital (rural, remote or metropolitan),[22] and the patient being treated at a different facility to where the diagnosis was originally made.[23] Additionally, out-of-date CPGs that do not represent the latest evidence,[15] and a lack of resources or time to implement CPG recommendations[15 24] have been found to influence CPG adherence. Patient and clinician characteristics have also been associated with adherence to specific CPGs, such as: older patient age,[4 11 14 22 23 25-28] and gender,[16] presence of comorbidities,[14 16 25] patient private health insurance status,[14 23] the socioeconomic status of patients,[29 30] patient race,[30 31] tumour-specific caseload of clinicians,[6 32] and the level of clinician specialty.[22] It is unknown whether these factors are associated with cancer treatment CPG adherence in Australia, and if there are similar patterns across cancer streams.

Objectives

Successful implementation of CPGs requires a context-specific approach[33] to overcome barriers that are specific to the Australian cancer care context. The systematic review will:

- Determine the estimated rates of adherence to cancer treatment CPG recommendations in Australia.
- Identify factors associated with adherence or non-adherence to those CPGs in Australia, estimating the association between those factors and adherence.
- Identify if there are associations between cancer treatment CPG adherence and patient outcomes in Australia (where possible).

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Methods and analysis:

This systematic review will be conducted in accordance with the Preferred Reporting for Items for Systematic Reviews and Meta-Analyses (PRISMA) statement,[34] and the protocol has been developed according to the PRISMA-P (protocol) checklist.[35] The systematic review has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020222962.

Search strategy

Five databases will be searched for eligible title/abstracts, including Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science. The abstracts identified will be collated in EndNote, with duplicates removed. The abstracts will then be reviewed for eligibility according to preestablished inclusion and exclusion criteria. Eligible studies will report the rate of adherence in Australia to cancer treatment CPGs. Examples of eligible study types include randomised studies, quasi-experimental designs and observational studies (e.g., cohort studies, case-control studies), and mixed methods studies that report quantitative data separately.

It is envisaged that some studies will include experimental interventions designed to influence adherence to CPG recommendations. Baseline data from experimental studies can be considered for meta-analysis along with data from observational

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studies; comparison of pre- and post-intervention data within experimental studies may provide information about barriers and facilitators.

Inclusion criteria

Studies must include empirical research, investigating the rate of receipt of cancer treatment CPG adherent care in Australia, and factors associated with adherence to those CPGs. The review will include empirical research articles published in English, between 2000-2021, restricted to the care of patients in Australia.

Additional full text inclusion criteria: The full text of each study must reference the relevant CPGs, and the measurement of adherence must be clearly defined.

Exclusion criteria

Non-empirical research, including conference abstracts, editorials, and opinion pieces will be excluded, as well as research which has not been peer reviewed. Studies that do not include patients in Australia and studies focusing on diseases other than cancer will also be excluded. This review will exclude studies referring to adherence to cancer prevention CPGs (e.g., screening CPGs), and CPGs for cancer treatment-associated side effects (e.g., pain management CPGs, antiemetic prophylaxis CPGs, and psychosocial care-focused CPGs, etc.).

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Additional full text exclusion criteria: The full text of studies will be excluded if they do not identify the rate of receipt of CPG adherent care, and/or factors associated with adherence to those CPGs, or fail to clearly indicate the CPG/s to which they refer.

Screening

1. Five experienced reviewers will independently review the title abstracts, including a 1% shared set of title abstracts which will be jointly reviewed, blinded to other reviewer decisions, to estimate inter-rater reliability. Title abstracts that may meet inclusion criteria and do not clearly breach exclusion criteria will be selected for full text review, while any disagreements will be reviewed by the team with a goal of reaching consensus regarding inclusion.
2. The full texts of the studies selected during title/abstract review will be reviewed by multiple reviewers. Articles that meet the inclusion criteria and do not breach the exclusion criteria will be included in the review. The reason for exclusion of each reviewed article will be recorded on a data extraction template. The reference lists of included papers will also be reviewed for potentially relevant articles not identified by database searches.

Data extraction

1. For all included articles, a data abstraction template will be used to extract data. The template will extract: the citation; location of study; study design;

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sample size; data source; age range and gender of participants; cancer stream(s); cancer stage(s); description of CPG(s) being assessed; measure of adherence; study intervention (if relevant); description of factors that are associated with adherence or non-adherence, including the strength of association of correlates with adherence/non-adherence; potential confounders (if available); and any available patient outcome data such as risk ratios, hazard ratios, five-year survival rates, patient satisfaction, etc.

2. A 10% sample of included studies will be reviewed by a second reviewer to confirm accuracy of data extraction.

Risk of bias and strength of evidence assessment

The quality appraisal and assessment of risk of bias for each included study will be conducted by two experienced reviewers. The Joanna Briggs Institute checklists for each study type (e.g., randomised control trials, cross sectional studies, cohort studies, or case control studies) will be used to assess the quality and risk of bias of each study. This data will be presented in a table.

Strategy for data synthesis

For each study, the frequency or rate of receipt of CPG adherent (or non-adherent) cancer treatment will be described. The factors associated with CPG adherent cancer treatment, including the strength of association, if available, will be described.

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The categorisation of these factors will be guided by the World Health Organisation's (WHO's) five dimensions of adherence framework, which includes: patient factors, socioeconomic factors, health condition factors, medical therapy factors, as well as factors related to the health care system and team.[36]

Where available, patient outcome rates (e.g., hazard ratios, five-year survival rates) of patients in receipt of CPG adherent treatment, compared to those in receipt of CPG non-adherent treatment, will be described. A narrative synthesis of qualitative data regarding adherence rates or factors related to CPG adherence, from included mixed methods studies, will be produced, if available. When two studies report adherence rates for a particular treatment, a pooled estimate of the rate of adherence (mean adherence) will be calculated by conducting a random effects meta-analysis, if appropriate. If non-adherence rates are reported, these will first be converted to adherence rates. Prior to pooling, forest plots, Q-tests, or I^2 tests will be used to assess heterogeneity.[37] Funnel plots will be used to identify publication bias.

Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research, and will not be involved in the research outlined by this protocol.

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Ethics and dissemination:

Ethics approval will not be required as this study will present results from other published studies. Upon completion of the systematic review, results will be published in a peer-reviewed journal, and as part of a dissertation (MB). Results may also be presented at international and national conferences, as well as forming part of academic presentations.

Authors' contributions:

MB was responsible for the conception of the study plan, and preparation of the manuscript.

JB, FR, GA, YT, BNGE, KL reviewed the study plan and manuscript and offered comments and edits.

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Competing interests statement.

No competing interests.

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PRISMA-P 2015 checklist: recommended items to include in a systematic review protocol

Moher D., Shamseer L., Clarke M., et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *The Journal of Systematic Reviews* 2015;4(1):1. doi: http://dx.doi.org/10.1186/2046-4053-4-1

Section/topic	Item #	Checklist item	Page/ Line
ADMINISTRATIVE INFORMATION			
Title			
Identification	1a	Identify the report as a protocol of a systematic review	P1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number	P2
Authors			
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	P1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	P9
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	
Support			
Sources	5a	Indicate sources of financial or other support for the review	P9
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	P4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	P5
METHODS			
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	P6

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Section/topic	Item #	Checklist item	Page/ Line
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	P5
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	P5
Study records			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	P7
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	P6-8
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	P5
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	P7
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	P7-8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	P7
Data			
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	P7-8
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	P8
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	P8
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	P8
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	P8
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	n/a

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BMJ Open

Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

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Title

Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

Corresponding author

Mia Bierbaum, mia.bierbaum@hdr.mq.edu.au,

Australian Institute of Health Innovation, 75 Talavera Rd, Macquarie University, NSW, Australia

Authors

[1] Bierbaum, Mia

[1] Rapport, Frances

[1,2] Arnolda, Gaston

[1,2] Tran, Yvonne

[1,2] Nic Giolla Easpaig, Brona

[1,3] Ludlow, Kristiana

[1,2] Braithwaite, Jeffrey

Affiliations

[1] Australian Institute of Heath Innovation, Macquarie University, NSW, Australia

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[2] Centre for Research Excellence in Implementation Science in Oncology, AIHI, Macquarie University, Sydney, Australia

[3] School of Psychology, University of Queensland, QLD, Australia

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Abstract:

Introduction: Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making. CPG adherent care is associated with improved patient survival outcomes; however, adherence rates are low across some cancer streams in Australia. Greater understanding of specific barriers to cancer treatment CPG adherence is warranted to inform future implementation strategies.

This paper presents the protocol for a systematic review that aims to determine cancer treatment CPG adherence rates in Australia across a variety of common

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cancers, and to identify any factors associated with adherence to those CPGs, as well as any associations between CPG adherence and patient outcomes.

Methods and analysis: Five databases will be searched, Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science, for eligible studies evaluating adherence rates to cancer treatment CPGs in Australia. A team of reviewers will screen the abstracts in pairs according to predetermined inclusion criteria and then review the full text of eligible studies. All included studies will be assessed for quality and risk of bias. Data will be extracted using a predefined data extraction template. The frequency or rate of adherence to CPGs, factors associated with adherence to those CPGs, and any reported patient outcome rates (e.g., relative risk ratios or 5-year survival rates) associated with adherence to CPGs, will be described. If applicable, a pooled estimate of the rate of adherence will be calculated by conducting a random effects meta-analysis. The systematic review will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

Ethics and dissemination: Ethics approval will not be required, as this review will present anonymised data from other published studies. Results from this study will form part of a doctoral dissertation (MB), will be published in a journal, presented at conferences, and other academic presentations.

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Strengths and limitations of this study

- This systematic review protocol describes a planned review that will use an internationally recognised methodology for the collection, extraction, and synthesis of data, enabling transparent presentation of methods and results to enable replication.
- The reporting of the systematic review will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.
- Abstracts and full texts will be reviewed by pairs of reviewers, with the lead reviewer reviewing all abstracts and full texts, and inter-rater reliability scores will be calculated for both screening stages.
- The main limitation of the review will be the exclusion of studies published in languages other than English, and the exclusion of non-empirical research, including conference abstracts, editorials, opinion pieces, and research which has not been peer reviewed.
- This study will also be limited to cancer treatment specific CPGs, excluding CPGs that relate to other elements of the cancer care journey such as pain management, and psychosocial care.

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Introduction

Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making,[1] and are designed to reduce clinical variation, through standardising clinical practice in line with best evidence, to ultimately enhance clinical outcomes.[2 3] There is growing evidence that CPG adherent cancer care is linked to improved patient survival rates.[4-8] Despite this, adherence to cancer treatment CPGs in Australia is often reported to be low across a variety of cancer streams.[4 6 9-14] Variation from CPG recommendations in some instances can be reasonably justified to cater for individual patient preferences and needs.[5 15] However, little is known about the factors that influence these decisions.[16] Low uptake and adherence to cancer treatment CPGs may also reflect poor quality CPGs, or poor implementation of CPGs. Greater understanding of specific barriers to adherence to cancer treatment CPGs is warranted, to inform future implementation strategies.

There are a multitude of internal (attitudinal) and external (organisational and structural) barriers to cancer treatment CPG adherence. These include clinician concern that CPGs are biased or oversimplified,[15 17] a lack of clinician awareness of, or agreement with, CPG recommendations,[15 17] inertia,[15 17] and concerns about the side effects of CPG recommended treatment.[15] Similarly, patient concerns about side effects or discomfort related to treatments, and access to treatment, have also been identified as barriers to CPG adherence.[18] Collaboration within multidisciplinary

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teams during treatment decision-making has been found to influence adherence to CPG recommendations,[19-21] as has patient access to services,[4] the location of the treating hospital (rural, remote or metropolitan),[22] and the patient being treated at a different facility to where the diagnosis was originally made.[23] Additionally, out-of-date CPGs that do not represent the latest evidence,[15] and a lack of resources or time to implement CPG recommendations[15 24] have been found to influence CPG adherence. Patient and clinician characteristics have also been associated with adherence to specific CPGs, such as: older patient age,[4 11 14 22 23 25-28] and gender,[16] presence of comorbidities,[14 16 25] patient private health insurance status,[14 23] the socioeconomic status of patients,[29 30] patient race,[30 31] tumour-specific caseload of clinicians,[6 32] and the level of clinician specialty.[22]

It is unknown whether these factors are associated with cancer treatment CPG adherence in Australia, and if there are similar patterns across cancer streams. Successful implementation of CPGs needs to be context-specific.[2 33] Therefore, the identification of factors specific to the Australian cancer treatment context is warranted in order to enable future CPG development, implementation and dissemination to be tailored according to identified facilitators and barriers of adherence.

Objectives

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The primary objective of the systematic review will be to:

- Determine the estimated rates of adherence to cancer treatment CPG recommendations in Australia.

The secondary objectives of the review will be to:

- Identify factors associated with adherence or non-adherence to cancer treatment CPGs in Australia, estimating the association between those factors and adherence.
- Identify if there are associations between cancer treatment CPG adherence and patient outcomes in Australia (where possible).

Methods and analysis:

This systematic review will be conducted in accordance with the Preferred Reporting for Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement,[34] and the protocol has been developed according to the PRISMA-P (protocol) checklist (online supplementary file 1).[35] The systematic review has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020222962. The systematic review database search began in March 2021, and it is anticipated that the review will be completed by March 2022. The Prospero registration will be updated if significant amendments are made to the protocol.

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Search strategy

Five databases will be searched for eligible title/abstracts, including Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science (please see online Supplementary File 2 for the detailed example of the search strategy applied in Embase). The abstracts identified will be collated in EndNote, with duplicates removed. The abstracts will then be reviewed for eligibility according to preestablished inclusion and exclusion criteria. Eligible studies will report the rate of adherence in Australia to cancer treatment CPGs. Examples of eligible study types include randomised studies, quasi-experimental designs and observational studies (e.g., cohort studies, case-control studies), and mixed methods studies that report quantitative data separately.

It is envisaged that some studies will include experimental interventions designed to influence adherence to CPG recommendations. Baseline data from experimental studies will be considered for meta-analysis along with data from observational studies; comparison of pre- and post-intervention data within experimental studies may provide information about barriers and facilitators.

Inclusion criteria

Studies must include empirical research, investigating the rate of receipt of cancer treatment CPG adherent care in Australia (including neoadjuvant, principal and adjuvant treatments), and factors associated with adherence to those CPGs. The

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review will include empirical research articles published in English, restricted to patient care within Australia.

Additional full text inclusion criteria: The full text of each study must reference the relevant CPGs, and clearly describe how CPG adherence was measured.

Exclusion criteria

Non-empirical research, including conference abstracts, editorials, and opinion pieces will be excluded, as well as research which has not been peer reviewed. Qualitative studies will be excluded. Studies that do not include patient care within Australia and studies focusing on diseases other than cancer will also be excluded. Studies that report data from both Australia and other countries will be excluded if the Australian data is not reported separately. This review will exclude studies referring to adherence to cancer prevention CPGs (e.g., screening CPGs), and CPGs for cancer treatment-associated side effects (e.g., pain management CPGs, antiemetic prophylaxis CPGs, and psychosocial care-focused CPGs, etc.).

Additional full text exclusion criteria: The full text of studies will be excluded if they do not identify the rate of receipt of CPG adherent care, and/or factors associated with adherence to those CPGs or fail to clearly indicate the CPG/s to which they refer.

Screening

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1. A team of reviewers, experienced in conducting systematic reviews, will conduct the title abstract screen in pairs. To enhance consistency of inclusion and exclusion of abstracts, a 10% sample of title abstracts will be reviewed by the whole team and discussed. The remaining abstracts will be divided amongst the reviewers, with the lead author reviewing every abstract in conjunction with another reviewer. Each reviewer will be blinded to the lead author's decisions; disagreements will be resolved by reviewer pairs reaching a consensus. Inter-rater reliability will be assessed between the lead author and each of the additional reviewers. Title abstracts that meet inclusion criteria and do not clearly breach exclusion criteria will be selected for full text review.
2. The full texts of the studies selected during the title abstract screen will be reviewed by the team in pairs. As in the abstract screening step, to enhance consistency of assessment of full texts, a 10% sample of full texts will be independently reviewed by the team of reviewers blinded to other reviewer decisions. The remaining full texts will be divided amongst the team of reviewers, with the lead author reviewing every full text in conjunction with another reviewer. Disagreements will be resolved by reviewer pairs reaching a consensus. Inter-rater reliability between the lead author and each reviewer will be calculated for the full text screening stage. All inter-rater reliability scores will be reported in the systematic review manuscript. Articles that meet the inclusion criteria and do not breach the exclusion criteria will be included

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in the review. The reason for exclusion of each reviewed article will be recorded on a data extraction template in Microsoft Excel. The reference lists of included papers will also be reviewed for potentially relevant articles not identified by database searches.

Data extraction

1. For all included articles, a data extraction template will be used to extract data. This template will be piloted by two reviewers on five full texts. Disagreements will be resolved through team consensus, and the tool will be revised if necessary. The template will extract: the citation; location of study; study design; sample size; data source; age range, gender and race (if applicable) of participants; cancer stream(s); cancer stage(s); description of CPG(s) being assessed; measure of adherence; study intervention (if relevant); description of factors reported in studies that are associated with CPG adherence or non-adherence (for example, patient age, Charlson Comorbidity Index, Socioeconomic Status, geographic remoteness, Country of Birth, Eastern Cooperative Oncology Group (ECOG) performance status, Aboriginal and Torres Strait Islander status, clinician case load, or hospital case load), including the strength of association of correlates with adherence/non-adherence; potential confounders (if available) and; any available patient outcome data such as risk ratios, hazard ratios, five-year

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survival rates, patient satisfaction, etc. All cancer streams will be included in the review.

2. All included studies will be checked by a second reviewer to confirm accuracy of data extraction. Any disagreements will be resolved through team consensus.

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Risk of bias and strength of evidence assessment

The quality appraisal and assessment of risk of bias for each included study will be conducted by two reviewers, experienced in conducting systematic reviews. The Joanna Briggs Institute checklists[36] for each study type (e.g., randomised control trials, cross sectional studies, cohort studies, or case control studies) will be used to assess the quality and risk of bias of each study, noting that questions differ for assessment of each study type. For example, the checklist questions for assessment of cross-sectional studies include, "*1. Were the criteria for inclusion in the sample clearly defined? 2. Were the study subjects and the setting described in detail? 3. Was the exposure measured in a valid and reliable way? 4. Were objective, standard criteria used for measurement of the condition? 5. Were confounding factors identified? 6. Were strategies to deal with confounding factors stated? 7. Were the outcomes measured in a valid and reliable way? 8. Was appropriate statistical analysis used?*".[36] These data will be reported in the manuscript. Each risk of bias assessment will be checked by a second reviewer. Any disagreements will be resolved through team consensus.

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Strategy for data synthesis

For each study, the frequency or rate of receipt of CPG adherent (or non-adherent) cancer treatment will be described. The factors associated with CPG adherent cancer treatment, including the strength of association, if available, will be described. The review will focus on variables that are significantly associated with CPG adherence in one or more studies. The categorisation of these variables will be guided by the World Health Organization's (WHO's) five dimensions of adherence framework, which includes: patient factors, socioeconomic factors, health condition factors, medical therapy factors, as well as factors related to the health care system and team.[37]

Where available, patient outcome rates (e.g., hazard ratios, five-year survival rates) of patients in receipt of CPG adherent treatment, compared to those in receipt of CPG non-adherent treatment, will be described. A narrative synthesis of qualitative data regarding adherence rates or factors related to CPG adherence, from included mixed methods studies, will be produced, if available. When two studies report adherence rates for a particular treatment, a pooled estimate of the rate of adherence (mean adherence) will be calculated by conducting a random effects meta-analysis, if appropriate. If non-adherence rates are reported, these will first be

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converted to adherence rates. Prior to pooling, forest plots, Q-tests, or I^2 tests will be used to assess heterogeneity.[38] Funnel plots will be used to identify publication bias.

Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research, and will not be involved in the research outlined by this protocol.

Ethics and dissemination:

Ethics approval will not be required as this study will only present anonymised data from other published studies. Upon completion of the systematic review, results will be published in a peer-reviewed journal, and as part of a dissertation (MB). Results may also be presented at international and national conferences, as well as other academic presentations. To translate the research into action, the findings from this work will be distributed to peak guideline development bodies, clinical societies involved in cancer treatment in Australia, and stakeholders involved in policy development and implementation in oncology.

Authors' contributions:

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MB was responsible for the conception of the study plan, and preparation of the manuscript.

JB, FR, GA, YT, BNGE, KL reviewed the study plan and manuscript and offered comments and edits.

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Competing interests statement.

No competing interests.

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Supplementary file 1: PRISMA-P 2015 checklist

Moher D., Shamseer L., Clarke M., et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *The Journal of Systematic Reviews* 2015;4(1):1. doi: http://dx.doi.org/10.1186/2046-4053-4-1

Section/topic	Item #	Checklist item	Page/ Line
ADMINISTRATIVE INFORMATION			
Title			
Identification	1a	Identify the report as a protocol of a systematic review	P1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number	P2
Authors			
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	P1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	P10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	P5
Support			
Sources	5a	Indicate sources of financial or other support for the review	P11
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	P4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	P5
METHODS			
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	P6

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Section/topic	Item #	Checklist item	Page/ Line
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	P6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	P6
Study records			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	P8
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	P7-9
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	P8
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	P8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	P8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	P8-9
Data			
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	P9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	P9-10
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	P9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	P9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	P8
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	n/a

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Supplementary file 2: Search strategy applied to Embase

1. practice guideline/
2. guideline*.ti,ab.
3. clinical protocol/
4. non-adherence.ti,ab.
5. non-compliance.ti,ab.
6. utili?ation.ti,ab.
7. underutili?ation.ti,ab.
8. Practice pattern*.ti,ab.
9. complian*.ti,ab.
10. adheren*.ti,ab.
11. concordan*.ti,ab.
12. malignant neoplasm/
13. cancer/
14. neoplasm/
15. carcinoma/
16. malignan*.ti,ab.
17. colorectal cancer/
18. lung tumor/
19. lung cancer/
20. breast cancer/
21. breast tumor/
22. prostate cancer/
23. melanoma/
24. 1 or 2 or 3
25. 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11
26. 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23
27. 24 and 25 and 26
28. limit 33 to (English language)
29. limit 28 to conference abstract
30. 28 not 29

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BMJ Open

Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

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Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, ONCOLOGY, PUBLIC HEALTH

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Title

Adherence to Clinical Practice Guidelines (CPGs) for the treatment of cancers in Australia and the factors associated with adherence: A systematic review protocol

Corresponding author

Mia Bierbaum, mia.bierbaum@hdr.mq.edu.au,

Australian Institute of Health Innovation, 75 Talavera Rd, Macquarie University, NSW, Australia

Authors

[1] Bierbaum, Mia

[1] Rapport, Frances

[1,2] Arnolda, Gaston

[1,2] Tran, Yvonne

[1,2] Nic Giolla Easpaig, Brona

[1,3] Ludlow, Kristiana

[1,2] Braithwaite, Jeffrey

Affiliations

[1] Australian Institute of Heath Innovation, Macquarie University, NSW, Australia

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[2] Centre for Research Excellence in Implementation Science in Oncology, AIHI, Macquarie University, Sydney, Australia

[3] School of Psychology, University of Queensland, QLD, Australia

Word Count

2,006 words

Key words

- Oncology
- Clinical Practice Guideline adherence
- Implementation science

Abstract:

Introduction: Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making. CPG adherent care is associated with improved patient survival outcomes; however, adherence rates are low across some cancer streams in Australia. Greater understanding of specific barriers to cancer treatment CPG adherence is warranted to inform future implementation strategies.

This paper presents the protocol for a systematic review that aims to determine cancer treatment CPG adherence rates in Australia across a variety of common

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cancers, and to identify any factors associated with adherence to those CPGs, as well as any associations between CPG adherence and patient outcomes.

Methods and analysis: Five databases will be searched, Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science, for eligible studies evaluating adherence rates to cancer treatment CPGs in Australia. A team of reviewers will screen the abstracts in pairs according to predetermined inclusion criteria and then review the full text of eligible studies. All included studies will be assessed for quality and risk of bias. Data will be extracted using a predefined data extraction template. The frequency or rate of adherence to CPGs, factors associated with adherence to those CPGs, and any reported patient outcome rates (e.g., relative risk ratios or 5-year survival rates) associated with adherence to CPGs, will be described. If applicable, a pooled estimate of the rate of adherence will be calculated by conducting a random effects meta-analysis. The systematic review will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

Ethics and dissemination: Ethics approval will not be required, as this review will present anonymised data from other published studies. Results from this study will form part of a doctoral dissertation (MB), will be published in a journal, presented at conferences, and other academic presentations.

Prospero number: CRD42020222962

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Strengths and limitations of this study

- This systematic review protocol describes a planned review that will use an internationally recognised methodology for the collection, extraction, and synthesis of data, enabling transparent presentation of methods and results to enable replication.
- The reporting of the systematic review will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.
- Abstracts and full texts will be reviewed by pairs of reviewers, with the lead reviewer reviewing all abstracts and full texts, and inter-rater reliability scores will be calculated for both screening stages.
- The main limitation of the review will be the exclusion of studies published in languages other than English, and the exclusion of non-empirical research, including conference abstracts, editorials, opinion pieces, and research which has not been peer reviewed.
- This study will also be limited to cancer treatment specific CPGs, excluding CPGs that relate to other elements of the cancer care journey such as pain management, and psychosocial care.

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Introduction

Clinical Practice Guidelines (CPGs) synthesise the latest evidence to support clinical and patient decision-making,[1] and are designed to reduce clinical variation, through standardisation of clinical practice in line with best evidence, to ultimately enhance clinical outcomes.[2 3] There is growing evidence that CPG adherent cancer care is linked to improved patient survival rates.[4-8] Despite this, adherence to cancer treatment CPGs in Australia is often reported to be low across a variety of cancer streams.[4 6 9-14] Variation from CPG recommendations in some instances can be reasonably justified to account for individual patient characteristics and preferences.[5 15] However, little is known about the factors that influence these decisions.[16] Low uptake and adherence to cancer treatment CPGs may also reflect poor quality CPGs, or poor implementation of CPGs. Greater understanding of specific barriers to adherence to cancer treatment CPGs is warranted, to inform future implementation strategies.

There are a multitude of internal (attitudinal) and external (organisational and structural) barriers to cancer treatment CPG adherence. These include clinician concern that CPGs are biased or oversimplified,[15 17] a lack of clinician awareness of, or agreement with, CPG recommendations,[15 17] inertia,[15 17] and concerns about the side effects of CPG recommended treatment.[15] Similarly, patient concerns about side effects or discomfort related to therapy, and treatment access, have also been identified as barriers to CPG adherence.[18] Collaboration within multidisciplinary

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teams during treatment decision-making has been found to influence adherence to CPG recommendations,[19-21] as has patient access to services,[4] the location of the treating hospital (rural, remote or metropolitan),[22] as well as the patient receiving care at a different facility from the initial treatment centre.[23] Additionally, out-of-date CPGs that do not represent the latest evidence,[15] and a lack of resources or time to implement CPG recommendations[15 24] have been found to influence CPG adherence. Patient and clinician characteristics associated with CPG adherence include older patient age,[4 11 14 22 23 25-28] race,[29 30] gender,[16] comorbid conditions,[14 16 25] patient private health insurance[14 23] and socioeconomic status,[29 31] as well as clinician specialty[22] and tumour-specific caseload.[6 32]

It is unknown whether these factors are associated with cancer treatment CPG adherence in Australia, and if there are similar patterns across cancer streams. Successful implementation of CPGs needs to be context-specific.[2 33] Therefore, the identification of factors specific to the Australian cancer treatment context is warranted in order to enable future CPG development, implementation and dissemination to be tailored according to identified facilitators and barriers of adherence within the country.

Objectives

The primary objective of the systematic review will be to:

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- Determine the estimated rates of adherence to cancer treatment CPG recommendations in Australia.

The secondary objectives of the review will be to:

- Identify factors associated with adherence or non-adherence to cancer treatment CPGs in Australia, estimating the association between those factors and adherence.
- Identify if there are associations between cancer treatment CPG adherence and patient outcomes in Australia (where possible).

Methods and analysis:

This systematic review will be conducted in accordance with the Preferred Reporting for Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement,[34] and the protocol has been developed according to the PRISMA-P (protocol) checklist (online supplementary file 1).[35] The systematic review has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020222962. The systematic review database search began in March 2021, and it is anticipated that the review will be completed by March 2022. The Prospero registration will be updated if significant amendments are made to the protocol.

Search strategy

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Five databases will be searched for eligible title/abstracts, including Ovid Medline, PsychInfo, Embase, Scopus, and Web of Science (please see online Supplementary File 2 for the detailed example of the search strategy applied in Embase). The abstracts identified will be collated in EndNote, with duplicates removed. The abstracts will then be reviewed for eligibility according to preestablished inclusion and exclusion criteria. Eligible studies will report the rate of adherence in Australia to cancer treatment CPGs. Examples of eligible study types include randomised studies, quasi-experimental designs and observational studies (e.g., cohort studies, case-control studies), and mixed methods studies that report quantitative data separately.

It is envisaged that some studies will include experimental interventions designed to influence adherence to CPG recommendations. Baseline data from experimental studies will be considered for meta-analysis along with data from observational studies; comparison of pre- and post-intervention data within experimental studies may provide information about barriers and facilitators.

Inclusion criteria

Studies must include empirical research, investigating the rate of receipt of cancer treatment CPG adherent care in Australia (including primary treatment, and neoadjuvant and adjuvant treatments), and factors associated with adherence to those CPGs. The review will include empirical research articles published in English, restricted to patient care within Australia. No date restrictions will be applied.

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Additional full text inclusion criteria: The full text of each study must reference the relevant CPGs, and clearly describe how CPG adherence was measured.

Exclusion criteria

Conference abstracts, editorials, and opinion pieces as well as purely qualitative research will be excluded. Studies that do not include patient care within Australia and studies focusing on diseases other than cancer will also be excluded. Studies that report data from both Australia and other countries will be excluded if the Australian data is not reported separately. This review will exclude studies referring to adherence to cancer prevention CPGs (e.g., screening CPGs), and CPGs for cancer treatment-associated side effects (e.g., pain management CPGs, antiemetic prophylaxis CPGs, and psychosocial care-focused CPGs, etc.).

Additional full text exclusion criteria: The full text of studies will be excluded if they do not identify the rate of receipt of CPG adherent care, and/or factors associated with adherence to those CPGs or fail to clearly indicate the CPG/s to which they refer.

Screening

1. A team of reviewers, experienced in conducting systematic reviews, will conduct the title abstract screen in pairs. To enhance consistency of inclusion and exclusion of abstracts, a 10% sample of title abstracts will be reviewed by

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the whole team and discussed. The remaining abstracts will be divided amongst the reviewers, with the lead author reviewing every abstract in conjunction with another reviewer. Each reviewer will be blinded to the lead author's decisions; disagreements will be resolved by reviewer pairs reaching a consensus. Inter-rater reliability will be assessed between the lead author and each of the additional reviewers. Title abstracts that meet inclusion criteria and do not clearly breach exclusion criteria will be selected for full text review.

2. The full texts of the studies selected during the title abstract screen will be reviewed by the team in pairs. As in the abstract screening step, to enhance consistency of assessment of full texts, a 10% sample of full texts will be independently reviewed by the team of reviewers blinded to other reviewer decisions. The remaining full texts will be divided amongst the team of reviewers, with the lead author reviewing every full text in conjunction with another reviewer. Disagreements will be resolved by reviewer pairs reaching a consensus. Inter-rater reliability between the lead author and each reviewer will be calculated for the full text screening stage. All inter-rater reliability scores will be reported in the systematic review manuscript. Articles that meet the inclusion criteria and do not breach the exclusion criteria will be included in the review. The reason for exclusion of each reviewed article will be recorded on a data extraction template in Microsoft Excel. The reference lists

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of included papers will also be reviewed for potentially relevant articles not identified by database searches.

Data extraction

1. For all included articles, a data extraction template will be used to extract data. This template will be piloted by two reviewers on five full texts. Disagreements will be resolved through team consensus, and the tool will be revised if necessary. The template will extract: the citation; location of study; study design; sample size; data source; age range, gender and race (if applicable) of participants; cancer stream(s); cancer stage(s); description of CPG(s) being assessed; measure of adherence; study intervention (if relevant); description of factors reported in studies that are associated with CPG adherence or non-adherence (for example, patient age, Charlson Comorbidity Index, Socioeconomic Status, geographic remoteness, Country of Birth, Eastern Cooperative Oncology Group (ECOG) performance status, Aboriginal and Torres Strait Islander status, clinician case load, or hospital case load), including the strength of association of correlates with adherence/non-adherence; potential confounders (if available) and; any available patient outcome data such as risk ratios, hazard ratios, five-year survival rates, patient satisfaction, etc. All cancer streams will be included in the review.

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2. All included studies will be checked by a second reviewer to confirm accuracy of data extraction. Any disagreements will be resolved through team consensus.

Risk of bias and strength of evidence assessment

The quality appraisal and assessment of risk of bias for each included study will be conducted by two reviewers, experienced in conducting systematic reviews. The Joanna Briggs Institute checklists[36] for each study type (e.g., randomised control trials, cross sectional studies, cohort studies, or case control studies) will be used to assess the quality and risk of bias of each study, noting that questions differ for assessment of each study type. For example, the checklist questions for assessment of cross-sectional studies include, "*1. Were the criteria for inclusion in the sample clearly defined? 2. Were the study subjects and the setting described in detail? 3. Was the exposure measured in a valid and reliable way? 4. Were objective, standard criteria used for measurement of the condition? 5. Were confounding factors identified? 6. Were strategies to deal with confounding factors stated? 7. Were the outcomes measured in a valid and reliable way? 8. Was appropriate statistical analysis used?*".[36] These data will be reported in the manuscript. Each risk of bias assessment will be checked by a second reviewer. Any disagreements will be resolved through team consensus.

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Strategy for data synthesis

For each study, the frequency or rate of receipt of CPG adherent (or non-adherent) cancer treatment will be described. The factors associated with CPG adherent cancer treatment, including the strength of association, if available, will be described. The review will focus on variables that are significantly associated with CPG adherence in one or more studies. The categorisation of these variables will be guided by the World Health Organization's (WHO's) five dimensions of adherence framework, which includes: patient factors, socioeconomic factors, health condition factors, medical therapy factors, as well as factors related to the health care system and team.[37]

Where available, patient outcome rates (e.g., hazard ratios, five-year survival rates) of patients in receipt of CPG adherent treatment, compared to those in receipt of CPG non-adherent treatment, will be described. A narrative synthesis of qualitative data regarding adherence rates or factors related to CPG adherence, from included mixed methods studies, will be produced, if available. When two studies report adherence rates for a particular treatment, a pooled estimate of the rate of adherence (mean adherence) will be calculated by conducting a random effects meta-analysis, if appropriate. If non-adherence rates are reported, these will first be converted to adherence rates. Prior to pooling, forest plots, Q-tests, or I^2 tests will be used to assess heterogeneity.[38] Funnel plots will be used to identify publication bias.

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Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research, and will not be involved in the research outlined by this protocol.

Ethics and dissemination:

Ethics approval will not be required as this study will only present anonymised data from other published studies. Upon completion of the systematic review, results will be published in a peer-reviewed journal, and as part of a dissertation (MB). Results may also be presented at international and national conferences, as well as other academic presentations. To translate the research into action, the findings from this work will be distributed to guideline development bodies, clinical societies involved in cancer treatment in Australia, and stakeholders involved in policy development and implementation in oncology.

Authors' contributions:

MB was responsible for the conception of the study plan, and preparation of the manuscript.

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JB, FR, GA, YT, BNGE, KL reviewed the study plan and manuscript and offered comments and edits.

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Competing interests statement.

No competing interests.

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Supplementary file 1: PRISMA-P 2015 checklist

Moher D., Shamseer L., Clarke M., et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *The Journal of Systematic Reviews* 2015;4(1):1. doi: http://dx.doi.org/10.1186/2046-4053-4-1

Section/topic	Item #	Checklist item	Page/ Line
ADMINISTRATIVE INFORMATION			
Title			
Identification	1a	Identify the report as a protocol of a systematic review	P1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number	P2
Authors			
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	P1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	P10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	P5
Support			
Sources	5a	Indicate sources of financial or other support for the review	P11
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	P4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	P5
METHODS			
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	P6

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Section/topic	Item #	Checklist item	Page/ Line
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	P6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	P6
Study records			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	P8
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	P7-9
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	P8
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	P8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	P8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	P8-9
Data			
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	P9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	P9-10
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	P9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	P9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	P8
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	n/a

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Supplementary file 2: Search strategy applied to Embase

1. practice guideline/
2. guideline*.ti,ab.
3. clinical protocol/
4. non-adherence.ti,ab.
5. non-compliance.ti,ab.
6. utili?ation.ti,ab.
7. underutili?ation.ti,ab.
8. Practice pattern*.ti,ab.
9. complian*.ti,ab.
10. adheren*.ti,ab.
11. concordan*.ti,ab.
12. malignant neoplasm/
13. cancer/
14. neoplasm/
15. carcinoma/
16. malignan*.ti,ab.
17. colorectal cancer/
18. lung tumor/
19. lung cancer/
20. breast cancer/
21. breast tumor/
22. prostate cancer/
23. melanoma/
24. 1 or 2 or 3
25. 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11
26. 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23
27. 24 and 25 and 26
28. limit 33 to (English language)
29. limit 28 to conference abstract
30. 28 not 29

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